

BLUESTOCKS FINTECH RESEARCH

NIFTY 100 FINANCIAL INTELLIGENCE ANALYST & ENGINEERING GUIDE

Comprehensive Operations Manual for Financial Screening, Multi-Year Valuation, Machine Learning Clustering, Automated PDF Tearsheets, and High-Throughput REST APIs.

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Target Audience: Equity Analysts, Portfolio Managers, Data Engineers & API Consumers

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System Architecture Overview

The Nifty 100 Financial Intelligence Platform is engineered as an enterprise-grade equity analysis and quantitative research environment. It ingests raw multi-year financial statements across all 92 active Nifty 100 constituents, normalizes disparate accounting conventions across 11 sectors, computes 30+ fundamental KPIs, applies machine learning clustering, and serves the structured knowledge through a Streamlit Web Application and a high-performance FastAPI REST server.

Layer	Components & Technologies	Core Responsibility
Data Storage	SQLite 3 (output/nifty100.db) with 20 performance indexes	ACID-compliant storage of statements, ratios, prices and metadata
Analytics Core	Pandas, NumPy, Scikit-Learn, SciPy	KPI ratios, CAGR growth, KMeans clustering, percentile rankings
Reporting Engine	ReportLab, Matplotlib, Seaborn	Automated 2-page tearsheets, sector deep-dives, correlation heatmaps
API Gateway	FastAPI, Starlette, Uvicorn (Port 8000)	Sub-15ms REST API serving JSON and streaming binary PDF assets
Interactive UI	Streamlit Multi-Page App (Port 8501)	Dynamic screening, radar visualisations, and financial statement exploration

Streamlit Dashboard Navigation Guide

The Streamlit dashboard offers an intuitive, reactive web interface designed for research analysts. Accessible locally at <http://localhost:8501>, the interface is split into 9 specialized pages accessible via the left sidebar.

Screen Name	Primary Capabilities	Key Visualizations & Outputs
01_home.py Executive Overview	High-level market snapshot, Nifty 100 coverage, broad sector breakdowns, aggregate median ROE, P/E, and D/E metrics.	KPI summary cards, sector distribution bar chart, top 5 ROE leaders table.
02_profiles.py Company Profile	Deep-dive company tearsheet view. Select any company ticker to view multi-year financial statements, ratios, and summary metrics.	P&L; table, balance sheet, cash flows, DuPont ROE breakdown, CAGR growth charts.
03_screener.py Stock Screener	Multi-metric filtering engine across 10 KPI dimensions with preset investment strategies and customizable sliders.	Filtered results table, composite score ranking, direct CSV export button.
04_peers.py Peer Group Analysis	Compare companies within any of the 11 sector peer groups with relative percentile rankings.	8-Axis interactive Radar comparison chart vs sector average and benchmark.
05_trends.py Historical Trends	Longitudinal time-series analysis (FY2019 to FY2024) of operating margins, revenue compounding, and free cash flow generation.	Interactive Plotly line charts, CAGR acceleration markers, margin expansion bars.
06_sectors.py Sector Deep-Dive	Cross-sector benchmark comparison comparing median valuations, leverage ratios, and capital expenditure intensities.	Sector P/E vs ROE scatter plots, median leverage boxplots, constituent tables.
07_capital.py Capital Allocation	Analysis of cash conversion, free cash flow generation, CFO quality scores, and debt servicing safety margins.	CFO vs PAT reconciliation waterfall, CapEx reinvestment rate bar charts.
08_reports.py Report Center	Central repository to download generated PDF company tearsheets, sector reports, and portfolio summaries.	Instant PDF downloads, batch report generation trigger, file status matrix.
09_settings.py Settings & Audit	System diagnostics, database health metrics, cache invalidation, and data quality rule validation audit logs.	Table row counts, cache clear button, DQ violation tables, environment status.

DASHBOARD PRODUCTIVITY TIP: To quickly navigate between company profiles, use the search dropdown on 02_profiles.py. The application leverages Streamlit's `@st.cache_data` with a 10-minute TTL to ensure instantaneous tab switching.

Stock Screener Engine & Strategy Presets

The Stock Screener (03_screener.py) allows analysts to filter all 92 companies simultaneously across 10 fundamental metrics. The screener computes a composite rank score to identify the most compelling ideas.

10 Supported Filtering Dimensions

Metric Name	Default Range	Direction	Financial Rationale & Significance
Return on Equity (ROE)	0% to 100%	Min (Floor)	Measures profitability on shareholders' equity base.
Return on Capital Employed (ROCE)	0% to 100%	Min (Floor)	Measures operating efficiency of total capital allocated.
Debt-to-Equity (D/E)	0.0 to 5.0	Max (Ceiling)	Filters balance sheet solvency and financial leverage risk.
Price-to-Earnings (P/E)	0.0 to 150.0	Max (Ceiling)	Identifies valuation multiples relative to trailing earnings.
Price-to-Book (P/B)	0.0 to 50.0	Max (Ceiling)	Assesses market valuation against net asset value.
Net Profit Margin (NPM)	-20% to 50%	Min (Floor)	Ensures pricing power and bottom-line margin strength.
Current Ratio	0.0 to 10.0	Min (Floor)	Validates short-term liquidity and working capital coverage.
Revenue CAGR 5-Year	-10% to 50%	Min (Floor)	Ensures sustained multi-year top-line business growth.
Dividend Payout	0% to 100%	Min (Floor)	Evaluates cash returned to shareholders via dividends.
Earnings Yield	0% to 25%	Min (Floor)	Inverse P/E metric reflecting baseline earnings return.

Built-in Strategy Presets

Preset Strategy	Filter Configuration	Ideal Investment Objective
Quality Compounders	ROE >= 15%, D/E <= 0.5, Rev CAGR >= 10%	Long-term wealth creation in blue-chip market leaders.
Debt-Free Champions	D/E == 0.0, ROE >= 15%, NPM >= 12%	Zero solvency risk compounders resilient to rate cycles.
High-Growth Momentum	Rev CAGR >= 15%, PAT CAGR >= 15%	Rapidly scaling mid/large caps with high earnings momentum.
Deep Value / Safety Margin	P/E <= 25, FCF >= 100 Cr, D/E <= 1.0	Undervalued cash generators with substantial margin of safety.
Free Cash Flow Machines	FCF >= 1,000 Cr, ROE >= 12%	Cash-rich defensive businesses with strong capital return.

Company Financial Profiles & Statements

The Company Profile screen (02_profiles.py) provides complete financial transparency for every company across 6 fiscal years (FY2019 to FY2024). The interface combines normalized financial statements with calculated financial ratios.

1. Normalized Statement Structure

Disparate accounting disclosures have been harmonized into canonical financial tables in SQLite:

Statement	Key Extracted Line Items	Analytical Insights
Income Statement (P&L;)	Revenue (Net Sales), Operating Income, Total Expenses, EBITDA, EBIT, Interest Expense, Depreciation, Tax Expense, Net Income (PAT), EPS.	Operating margin trend, operating leverage, tax efficiency, and bottom-line earnings quality.
Balance Sheet	Total Equity, Share Capital, Reserves & Surplus, Total Debt (Long + Short Term), Total Assets, Current Assets, Current Liabilities, Net Fixed Assets.	Solvency, working capital intensity, net debt trajectory, and asset base expansion.
Cash Flow Statement	Operating Cash Flow (CFO), Capital Expenditure (CapEx), Investing Cash Flow (CFI), Financing Cash Flow (CFF), Free Cash Flow (FCF = CFO - CapEx).	Cash conversion ratio (CFO / PAT), organic FCF yield, dividend sustainability.
Valuation & Multiples	Market Capitalisation, Enterprise Value (EV), P/E Ratio, P/B Ratio, EV/EBITDA, Dividend Yield.	Historical multiple expansion/compression across market cycles.

2. DuPont ROE Decomposition Model

To diagnose whether high Return on Equity is driven by operating efficiency or balance sheet leverage, the profile screen evaluates the classic 3-Stage DuPont Decomposition formula:

ROE = (Net Profit Margin) × (Asset Turnover) × (Equity Multiplier)
= (Net Profit / Revenue) × (Revenue / Total Assets) × (Total Assets / Total Equity)

Interpretation:

- High NPM: Superior pricing power and cost control (e.g. IT & FMCG).
- High Asset Turnover: Efficient capital utilisation (e.g. Retail & FMCG).
- High Equity Multiplier: Leverage-driven ROE (e.g. Financials & Real Estate).

FINANCIAL ANALYSIS HEURISTIC: Look for companies with expanding ROE driven by Net Profit Margin or Asset Turnover rather than increasing Equity Multiplier (Debt). Leverage-driven ROE expansion carries heightened vulnerability during rate hikes.

Peer Group Analytics & 8-Axis Radar Charts

The Peer Comparison module (04_peers.py) allows relative valuation and operational benchmarking against sector competitors. Every company is benchmarked across 10 percentile-ranked dimensions.

8-Axis Radar Chart Interpretation

The 8-axis radar visualization provides a normalized 0-to-100 percentile profile comparing a target company against its peer group average and the designated sector benchmark company:

Radar Axis Metric	Metric Type	Higher Percentile Meaning
1. ROE (%)	Profitability	Top-tier return on equity relative to industry peers.
2. ROCE (%)	Capital Efficiency	Optimal operating earnings generated per unit of total capital.
3. Operating Profit Margin (OPM)	Core Efficiency	Widest gross operating spread before financing costs.
4. Net Profit Margin (NPM)	Bottom-Line Power	Resilient net earnings after all taxes and overheads.
5. Interest Coverage Ratio (ICR)	Solvency Buffer	Substantial earnings cushion to service debt obligations.
6. Asset Turnover	Asset Efficiency	Superior asset productivity generating top-line revenue.
7. Revenue CAGR 5-Year	Growth Velocity	Sector-leading compounding of annual revenue.
8. Low Leverage (1 / D/E)	Balance Sheet Health	Lowest relative debt-to-equity ratio in the peer group.

Peer Groups Across 11 Nifty Sectors

Companies are grouped into 11 canonical sector cohorts, each with a designated benchmark anchor:

Peer Group Name	Company Count	Sector Benchmark Anchor	Key Industry Constituents
Information Technology	5	TCS	TCS, INFY, HCLTECH, LTIM, TECHM
Financial Services	22	HDFCBANK	HDFCBANK, ICICIBANK, SBIN, KOTAKBANK, BAJFINANCE
Consumer Discretionary	15	TITAN	TITAN, MARUTI, TATAMOTORS, M&M;, BAJAJ-AUTO
Consumer Staples (FMCG)	12	HINDUNILVR	HINDUNILVR, ITC, NESTLEIND, BRITANNIA, TATACONSUM
Energy & Oil/Gas	8	RELIANCE	RELIANCE, ONGC, NTPC, POWERGRID, BPCL, IOC
Healthcare & Pharma	10	SUNPHARMA	SUNPHARMA, CIPLA, DRREDDY, DIVISLAB, APOLLOHOSP

Machine Learning Clustering & Outlier Profiling

To discover objective peer cohorts beyond conventional industrial sector classifications, the analytics engine (src/analytics/clustering.py) applies unsupervised KMeans clustering (k=5) and multi-dimensional Z-score outlier detection.

1. KMeans Clustering Pipeline

Clustering is executed on 5 standardized financial features across all 92 companies:

```
1. Return on Equity (%)           [return_on_equity_pct]
2. Debt-to-Equity                 [debt_to_equity]
3. Revenue 5Y CAGR (%)           [revenue_cagr_5yr]
4. Free Cash Flow 5Y CAGR (%)    [fcf_cagr_5yr]
5. Operating Profit Margin (%)    [operating_profit_margin_pct]
Methodology:
• Missing value imputation using sector median.
• Feature normalization using StandardScaler (zero mean, unit variance).
• KMeans execution with k=5, random_state=42 for deterministic clustering.
```

2. Cluster Profiles & Typology

Cluster ID & Name	Financial Characteristics	Representative Companies
Cluster 0 High-Quality Compounders	High ROE (>25%), low D/E (<0.3), strong OPM (>20%), stable double-digit CAGR.	TCS, INFY, TITAN, HINDUNILVR
Cluster 1 Defensive Dividend Payers	Moderate ROE (12-18%), healthy dividend payout, high cash conversion, low beta.	ITC, POWERGRID, NTPC, COALINDIA
Cluster 2 Emerging High Growth	Rapid revenue CAGR (>18%), high reinvestment rate, moderate margins.	TRENT, ZOMATO, BEL, HAL
Cluster 3 Leveraged / Capital Intensive	Higher debt-to-equity (>1.5), substantial CapEx, cyclical margins.	TATASTEEL, JSWSTEEL, ADANIENT
Cluster 4 Turnaround / Value Cyclical	Low valuation multiples, recovering margins, cash flow inflection.	TATAMOTORS, VEDL, HINDALCO

3. Multi-Metric Outlier Detection

Outliers are flagged when a company's metric deviates by more than 3 standard deviations ($|Z\text{-score}| > 3.0$) from its broad sector median. Outlier reports are generated at output/outlier_report.csv.

Automated PDF Reporting & Tearsheet Generation

The automated reporting engine (src/reports/) builds publication-ready PDF equity research reports with pixel-perfect vector graphics, DuPont analysis tables, and CAGR trajectory charts.

Report Catalog & Directory Layout

Report Type	Output Location	Page Count	Contents & Structure
Company Tearsheets	reports/tearsheets/{TICKER}.pdf	2 Pages / Co.	Page 1: Profile, KPIs, DuPont analysis, valuation. Page 2: 6-year P&L, balance sheet & cash flow history.
Sector Summaries	reports/sector/{SECTOR}.pdf	3-5 Pages	Sector median comparisons, constituent ranking, CapEx intensity, valuation scatter plots.
Portfolio Summary	reports/portfolio/portfolio_summary.pdf	92 Pages	One-page executive tear card per company in alphabetical order with trend directional arrows.
Correlation Matrix	reports/correlation_heatmap.png	Graphic Image	10x10 Pearson correlation heatmap across all KPIs.
Elbow Curve Plot	reports/elbow_plot.png	Graphic Image	Inertia vs k (2-10) validating optimal cluster count.

Batch Tearsheet Generation via CLI

Analysts can generate tearsheets on demand via Python scripts or API requests:

```
# 1. Generate tearsheet for single company (e.g. TCS)
python -c "from src.reports.tearsheet import generate_company_tearsheet; generate_company_tearsheet('TCS')"
# 2. Batch generate all 92 tearsheets
python src/reports/generate_all_tearsheets.py
# 3. Generate portfolio summary PDF
python src/reports/portfolio_summary.py
```

STREAMLINED REPORT ACCESS: All generated PDFs are automatically made available for instant one-click download in the Streamlit UI under the `08_reports.py` page or via the `/api/v1/companies/{ticker}/tearsheet` API endpoint.

FastAPI REST Reference & curl Examples

The platform exposes a high-throughput REST API powered by FastAPI on port 8000. Interactive Swagger documentation is available at <http://localhost:8000/docs>.

Endpoint Method & Path	Key Query Parameters	Response Summary & Status Codes
GET /api/v1/health	None	HTTP 200: {status: 'ok', db_row_counts: {...}, uptime: 120}
GET /api/v1/companies	sector, market_cap_category, search, limit	HTTP 200: Paginated list of 92 companies with sector and latest KPIs.
GET /api/v1/companies/{ticker}	ticker (path)	HTTP 200: Full profile with latest ratios, P&L, BS, CF. (HTTP 404 if not found)
GET /api/v1/companies/{ticker}/pi	from_year, to_year	HTTP 200: Multi-year income statement time series.
GET /api/v1/screener	min_roe, max_de, min_fcf, sector, max_pe	HTTP 200: Ranked screener result list. (HTTP 400 on invalid param bounds)
GET /api/v1/sectors	None	HTTP 200: 11 sectors with company counts, median ROE, PE, and D/E.
GET /api/v1/peers/{group_name}	group_name (path)	HTTP 200: Peer group constituent metrics and percentile ranks.
GET /api/v1/market-cap/{ticker}	ticker (path)	HTTP 200: 2019-2024 valuation history (P/E, P/B, EV/EBITDA, Div Yield).

Practical curl Command Recipes

```
# 1. Health Check
curl -X GET "http://localhost:8000/api/v1/health"

# 2. Screener: Quality Compounders (ROE >= 20%, D/E <= 0.5)
curl -X GET "http://localhost:8000/api/v1/screener?min_roe=20.0&max_de=0.5"

# 3. Company Financial Profile (TCS)
curl -X GET "http://localhost:8000/api/v1/companies/TCS"

# 4. Peer Radar Comparison (INFY)
curl -X GET "http://localhost:8000/api/v1/companies/INFY/peers/compare"

# 5. Download Binary PDF Tearsheet (RELIANCE)
curl -X GET "http://localhost:8000/api/v1/companies/RELIANCE/tearsheet" --output RELIANCE_tearsheet.pdf
```

Data Quality Framework & Integrity Governance

To ensure zero garbage in financial models, the ingestion pipeline runs an automated 16-rule Data Quality (DQ) validation suite (src/analytics/data_quality.py). Data anomalies are categorized by severity.

Rule ID	Rule Name & Description	Severity Tier	Corrective Action
DQ-01	Completeness: Mandatory financial statement fields must not be NULL.	CRITICAL	Quarantines row to output/dq_violations.csv
DQ-02	Ticker Uniqueness: No duplicate company tickers within a fiscal year.	CRITICAL	Deduplicates based on latest updated timestamp
DQ-03	Balance Sheet Balance: Total Assets == Total Liabilities + Equity.	CRITICAL	Flags accounting mismatch if difference > 1 Cr
DQ-04	Non-Negative Revenue: Net sales must be strictly > 0.	CRITICAL	Rejects negative revenue entries
DQ-05	Positive Share Capital: Issued equity capital must be > 0.	CRITICAL	Flags potential data corruption in share ledger
DQ-06	OPM Range Check: Operating margin must lie within [-100%, 100%].	WARNING	Emits warning log for manual analyst review
DQ-07	ROE Extreme Check: ROE > 150% flagged for high leverage verification.	WARNING	Verifies whether high ROE stems from thin equity
DQ-08	Negative Equity Handled: Prevents nonsensical positive ROE on negative net worth.	CRITICAL	Sets ROE to None and triggers distress flag
DQ-09	Zero Interest Coverage: Handles debt-free zero interest expense.	INFO	Sets ICR to None or safe sentinel without zero-division error
DQ-10	Non-Financial D/E Limit: Flags non-banking companies with D/E > 5.0.	WARNING	Triggers high financial leverage risk flag
DQ-11	CAGR Base Sign Check: Prevents invalid CAGR calculation on negative base year.	WARNING	Marks turnaround or loss-to-profit status
DQ-12	Cash Flow Reconciliation: CFO vs Operating Income reconciliation.	WARNING	Calculates CFO Quality Score (CFO / EBITDA)

DATA GOVERNANCE COMMAND: To re-run the full data quality validation suite against the active database, execute `pytest tests/dq/ -v`. All 15 automated DQ unit tests must pass with 0 failures.

Troubleshooting, Operations & FAQ

This section documents operational procedures, common troubleshooting scenarios, and solutions for running the Nifty 100 platform in development and staging environments.

Symptom / Error	Root Cause	Step-by-Step Resolution
sqlite3.OperationalError: database is locked	Multiple write processes attempting to access SQLite file simultaneously.	All API routers use read-only queries with <code>check_same_thread=False</code> . For writes, ensure background ETL runs sequentially.
Port 8000 / 8501 already in use	Previous Uvicorn or Streamlit instance was not cleanly terminated.	Kill process using port: <code>netstat -ano findstr :8000</code> <code>taskkill /PID {PID} /F</code>
HTTP 404: Ticker not found	Requested company ticker is not present in the 92 active Nifty 100 constituents.	Verify ticker against <code>GET /api/v1/companies</code> . Tickers are case-insensitive and auto-capitalized.
HTTP 400: Invalid screener parameter	Filter parameter value was out of allowable bounds (e.g. <code>min_roe=150</code>).	Ensure <code>min_roe</code> in <code>[0, 100]</code> , <code>max_de</code> in <code>[0, 50]</code> , <code>max_pe</code> in <code>[0, 500]</code> . Refer to OpenAPI documentation.
ReportLab font missing	System fonts unavailable during PDF compilation.	The reporting engine uses standard built-in PostScript fonts (Helvetica, Courier) for universal portability.

System Startup & Operations Checklist

```
# 1. Activate Python Virtual Environment
.\.venv\Scripts\activate
# 2. Run Full Automated Test Suite (667 tests)
pytest tests/ --html=reports/pytest_report.html --self-contained-html
# 3. Start FastAPI REST Backend (Port 8000)
uvicorn src.api.main:app --host 0.0.0.0 --port 8000 --reload
# 4. Start Streamlit Frontend Dashboard (Port 8501)
streamlit run src/dashboard/app.py --server.port 8501
```

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